

torch.isclose#

<https://pytorch.org/docs/stable/generated/torch.isclose.html>

Description:

Returns a new tensor with boolean elements representing if each element of input is “close” to the corresponding element of other. Closeness is defined as: where input and other are finite. Where input and/or other are nonfinite they are close if and only if they are equal, with NaNs being considered equal to each other when equal_nan is True. input (Tensor) – first tensor to compare other (Tensor) – second tensor to compare rtol (float, optional) – relative tolerance. Default: 1e-05

Function Signature

```
torch.isclose(input, other, rtol=1e-05, atol=1e-08,  
equal_nan=False) → Tensor#
```

Parameters

Code Examples

```
>>> torch.isclose(torch.tensor((1., 2, 3)), torch.tensor((1 +  
1e-10, 3, 4)))  
tensor([ True, False, False])  
>>> torch.isclose(torch.tensor((float('inf'), 4)),  
torch.tensor((float('inf'), 6)), rtol=.5)  
tensor([True, True])
```

Detailed Documentation

Documentation

Returns a new tensor with boolean elements representing if each element of input is “close” to the corresponding element of other. Closeness is defined as:

where input and other are finite. Where input and/or other are nonfinite they are close if and only if they are equal, with NaNs being considered equal to each other when equal_nan is True.

input (Tensor) – first tensor to compare

other (Tensor) – second tensor to compare

rtol (float, optional) – relative tolerance. Default: 1e-05

atol (float, optional) – absolute tolerance. Default: 1e-08

equal_nan (bool, optional) – if True, then two NaN s will be considered equal. Default: False